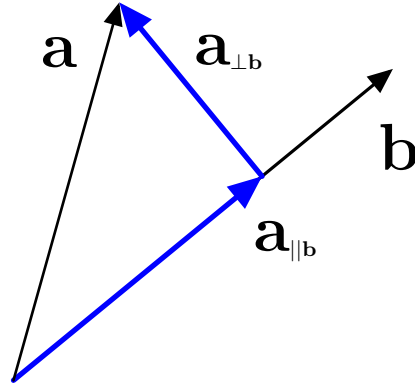


In the last essay we developed the tools needed to multiply bivectors. In this paper we will take a deeper look at how bivectors can be used to measure the separation between lines. As applications of our hard work, exercises regarding the existence of certain configurations are given, and we will prove a couple of Miquel's theorems regarding cyclic quadrilaterals.

Previously we developed the tools to claim that

$$\begin{aligned} (\mathbf{a} \wedge \mathbf{b})(\mathbf{c} \wedge \mathbf{d}) = & (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c}) - (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) \\ & + (\mathbf{b} \cdot \mathbf{c})\mathbf{a} \wedge \mathbf{d} - (\mathbf{b} \cdot \mathbf{d})\mathbf{a} \wedge \mathbf{c} \\ & + (\mathbf{a} \cdot \mathbf{d})\mathbf{b} \wedge \mathbf{c} - (\mathbf{a} \cdot \mathbf{c})\mathbf{b} \wedge \mathbf{d} \\ & + \mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c} \wedge \mathbf{d} \end{aligned}$$

which a quick substitution shows that when $\mathbf{a} = x_1\mathbf{e}_1 + y_1\mathbf{e}_2$, $\mathbf{b} = x_2\mathbf{e}_1 + y_2\mathbf{e}_2$, $\mathbf{c} = x_3\mathbf{e}_1 + y_3\mathbf{e}_2$, and $\mathbf{d} = x_4\mathbf{e}_1 + y_4\mathbf{e}_2$ our vectors lie in a plane we have $(\mathbf{a} \wedge \mathbf{b})(\mathbf{c} \wedge \mathbf{d}) = (\mathbf{a} \wedge \mathbf{b}) * (\mathbf{c} \wedge \mathbf{d})$ only the scalar part survives of their product. This means that we actually have a necessary and sufficient condition for two bivectors to be representatives of the same space: their product is a scalar. In this paper we will be exclusively concerned with bivectors in the plane. As usual we will want to know that geometric algebra makes it easy to compute projections $\mathbf{a}_{\parallel\mathbf{b}} = (\mathbf{a} \cdot \mathbf{b})\mathbf{b}^{-1}$ and rejections $\mathbf{a}_{\perp\mathbf{b}} = (\mathbf{a} \wedge \mathbf{b})\mathbf{b}^{-1}$.



We saw in a previous essay how to define two different unoriented measurements of the separation between two non-null vectors called spread and cross

$$\begin{aligned} s(\mathbf{a}, \mathbf{b}) &= \frac{\mathbf{a}_{\perp\mathbf{b}}^2}{\mathbf{a}^2} = \frac{-(\mathbf{a} \wedge \mathbf{b})^2}{\mathbf{a}^2\mathbf{b}^2} = \frac{\mathbf{b}_{\perp\mathbf{a}}^2}{\mathbf{b}^2} = s(\mathbf{b}, \mathbf{a}) \\ c(\mathbf{a}, \mathbf{b}) &= \frac{\mathbf{a}_{\parallel\mathbf{b}}^2}{\mathbf{a}^2} = \frac{(\mathbf{a} \cdot \mathbf{b})^2}{\mathbf{a}^2\mathbf{b}^2} = \frac{\mathbf{b}_{\parallel\mathbf{a}}^2}{\mathbf{b}^2} = c(\mathbf{b}, \mathbf{a}) \end{aligned}$$

which by either the Pythagorean theorem or the fundamental identity of the geometric product can be seen to satisfy $s(\mathbf{a}, \mathbf{b}) + c(\mathbf{a}, \mathbf{b}) = 1$.

We also saw that the three spreads and crosses of a triangle both satisfy cubic relations which become quadratic in one value if the other two are fixed. This means that two opposite spreads in a quadrilateral being equal is not sufficient to ensure that the quadrilateral is cyclic.

To overcome these pitfalls, we will introduce two more measurements between vectors, but this time they are oriented. Given two non perpendicular but otherwise possibly null vectors $\mathbf{a}$ and $\mathbf{b}$, we define their **turn** to be the purely bivector quantity

$$u(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}_{\perp \mathbf{b}}}{\mathbf{a}_{\parallel \mathbf{b}}} = \frac{\mathbf{a} \wedge \mathbf{b}}{\mathbf{a} \cdot \mathbf{b}} = \frac{-\mathbf{b} \wedge \mathbf{a}}{\mathbf{b} \cdot \mathbf{a}} = \frac{-\mathbf{b}_{\perp \mathbf{a}}}{\mathbf{b}_{\parallel \mathbf{a}}} = -u(\mathbf{b}, \mathbf{a})$$

which is zero exactly when $\mathbf{a}$ and $\mathbf{b}$ are parallel. Next we define the **coturn** of two non parallel vectors to be the reciprocal of their turn $o(\mathbf{a}, \mathbf{b}) = \frac{1}{u(\mathbf{a}, \mathbf{b})} = \frac{\mathbf{a}_{\parallel \mathbf{b}}}{\mathbf{a}_{\perp \mathbf{b}}} = -o(\mathbf{b}, \mathbf{a})$ which is zero precisely when $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

Theorem 1 Suppose we are given a triangle ABC and we give it an oriented boundary $\mathbf{a} = \mathbf{C} - \mathbf{B}$, $\mathbf{b} = \mathbf{A} - \mathbf{C}$, and $\mathbf{c} = \mathbf{B} - \mathbf{A}$, so that $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$ then we have

$$\begin{aligned} u(\mathbf{a}, \mathbf{b}) + u(\mathbf{b}, \mathbf{c}) + u(\mathbf{c}, \mathbf{a}) &= -u(\mathbf{a}, \mathbf{b})u(\mathbf{b}, \mathbf{c})u(\mathbf{c}, \mathbf{a}) \\ o(\mathbf{a}, \mathbf{b})o(\mathbf{b}, \mathbf{c}) + o(\mathbf{b}, \mathbf{c})o(\mathbf{c}, \mathbf{a}) + o(\mathbf{c}, \mathbf{a})o(\mathbf{a}, \mathbf{b}) &= -1 \end{aligned}$$

whenever either triplet of turns or coturns is defined.

Proof We will prove the first equation, the second follows from the first equation by dividing both sides by the quantity $u(\mathbf{a}, \mathbf{b})u(\mathbf{b}, \mathbf{c})u(\mathbf{c}, \mathbf{a})$, although the second equation can also be proved first in a similar fashion to what we do next. We will not be particularly clever or geometrically inspired with this proof. The left hand side of the first equation is

$$u(\mathbf{a}, \mathbf{b}) + u(\mathbf{b}, \mathbf{c}) + u(\mathbf{c}, \mathbf{a}) = \frac{\mathbf{a} \wedge \mathbf{b}}{\mathbf{a} \cdot \mathbf{b}} + \frac{\mathbf{b} \wedge \mathbf{c}}{\mathbf{b} \cdot \mathbf{c}} + \frac{\mathbf{c} \wedge \mathbf{a}}{\mathbf{c} \cdot \mathbf{a}}$$

which becomes

$$u(\mathbf{a}, \mathbf{b}) + u(\mathbf{b}, \mathbf{c}) + u(\mathbf{c}, \mathbf{a}) = \frac{(\mathbf{b} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{a})\mathbf{a} \wedge \mathbf{b} + (\mathbf{c} \cdot \mathbf{a})(\mathbf{a} \cdot \mathbf{b})\mathbf{b} \wedge \mathbf{c} + (\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c})\mathbf{c} \wedge \mathbf{a}}{(\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{a})}$$

after writing our terms with common denominators. Next, we want to use the fact that $\mathbf{c} = -\mathbf{a} - \mathbf{b}$ to say $\mathbf{b} \wedge \mathbf{c} = \mathbf{c} \wedge \mathbf{a} = -\mathbf{b} \wedge \mathbf{a} = \mathbf{a} \wedge \mathbf{b}$ so that

$$u(\mathbf{a}, \mathbf{b}) + u(\mathbf{b}, \mathbf{c}) + u(\mathbf{c}, \mathbf{a}) = \frac{\mathbf{a} \wedge \mathbf{b}((\mathbf{b} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{a}) + (\mathbf{c} \cdot \mathbf{a})(\mathbf{a} \cdot \mathbf{b}) + (\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c}))}{(\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{a})}.$$

Now we want to observe that $(\mathbf{b} \wedge \mathbf{c})(\mathbf{c} \wedge \mathbf{a}) = (\mathbf{a} \cdot \mathbf{b})\mathbf{c}^2 - (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{c})$ as well as

$$(\mathbf{b} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{a}) + (\mathbf{c} \cdot \mathbf{a})(\mathbf{a} \cdot \mathbf{b}) + (\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c}) = (\mathbf{b} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{a}) + (\mathbf{a} \cdot \mathbf{b})(\mathbf{c} \cdot (\mathbf{a} + \mathbf{b})).$$

Putting all this together with the fact that $\mathbf{c} = -\mathbf{a} - \mathbf{b}$ we get

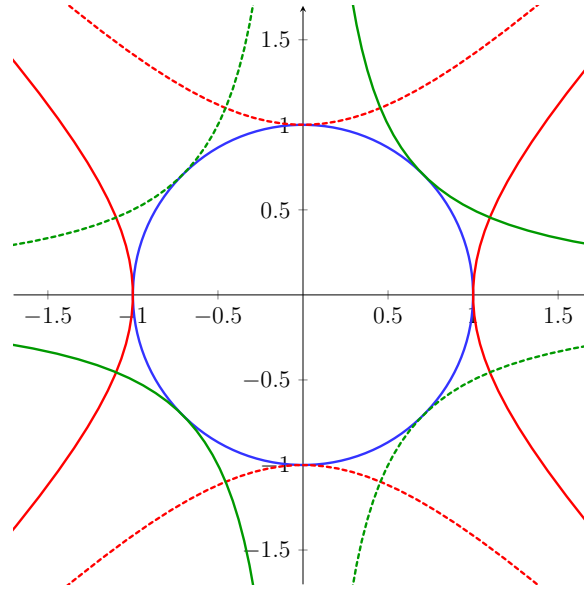
$$(\mathbf{b} \cdot \mathbf{c})(\mathbf{c} \cdot \mathbf{a}) + (\mathbf{c} \cdot \mathbf{a})(\mathbf{a} \cdot \mathbf{b}) + (\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c}) = -(\mathbf{b} \wedge \mathbf{c})(\mathbf{c} \wedge \mathbf{a})$$

as required.

Notice that we now have cubic relations which become linear if two of the three unknowns are fixed. From their definitions, we can see that turn and coturn are well defined, that is for scalars α and β we have $u(\alpha \mathbf{a}, \beta \mathbf{b}) = u(\mathbf{a}, \mathbf{b})$ and $o(\alpha \mathbf{a}, \beta \mathbf{b}) = o(\mathbf{a}, \mathbf{b})$. This means that our relations actually hold for any three vectors in a plane, regardless of whether they form the boundary of a triangle or not.

Exercise 1 In blue geometry with a basis $\mathbf{b}_1, \mathbf{b}_2$ where $\mathbf{b}_1^2 = \mathbf{b}_2^2 = 1$ and $\mathbf{b}_1 \perp \mathbf{b}_2$ we can represent vectors using polar coordinates $\mathbf{v}(r, \theta) = r \cos(\theta) \mathbf{b}_1 + r \sin(\theta) \mathbf{b}_2$ and oriented angles. Use this to show $u(\mathbf{v}(r_1, \theta_1), \mathbf{v}(r_2, \theta_2)) = \tan(\theta_2 - \theta_1) \mathbf{b}_1 \mathbf{b}_2$ and for angles $\alpha + \beta + \gamma = \pi$ that $\tan(\alpha) + \tan(\beta) + \tan(\gamma) = \tan(\alpha) \tan(\beta) \tan(\gamma)$. Explain the sign difference in the bivector identity compared to the blue angle identity. Find and state the corresponding identities in red geometry.

At this point we want to investigate applications of these types of measurements to configurations inscribed on circles. Recall that a **circle** with quadrance R^2 and center O is the equation $(X - O)^2 = R^2$ restricted to a plane. Pictured below are the blue, red, green unit circles and the red, green anti unit circles.



For convenience we usually perform a change of coordinates so that the center of our circle is the origin, and by a change of coordinates we can always leave that simpler case. As an abuse of notation, when we add or multiply points, we are really composing their position vectors. Despite seemingly being relegated to a secondary role, the coturn is as, if not more important than the turn. This is due to the fact that coturn plays better with circles.

Theorem 2 Given a triangle of distinct noncolinear points ABC inscribed on $X^2 = R^2$ a circle centered at the origin O , the coturn $o(C - B, C - A)$ is independent of C , it depends only on the quadrance of the circle and the segment $B - A$.

Proof Since triangle ABC lies in the plane determined by AOB , we can express $C = \alpha A + \beta B$ for some numbers α and β . Let us compute the numerator and denominator of $o(C-A, C-B)$ separately. First we have

$$(C - A) \cdot (C - B) = C^2 - C \cdot (A + B) + A \cdot B = C^2 - \alpha A^2 - \beta B^2 - (\alpha + \beta)A \cdot B + A \cdot B,$$

but using the fact that $A^2 = B^2 = C^2 = R^2$ we can simplify our numerator to

$$(C - A) \cdot (C - B) = (1 - \alpha - \beta)(A \cdot B + R^2).$$

Next up is our denominator

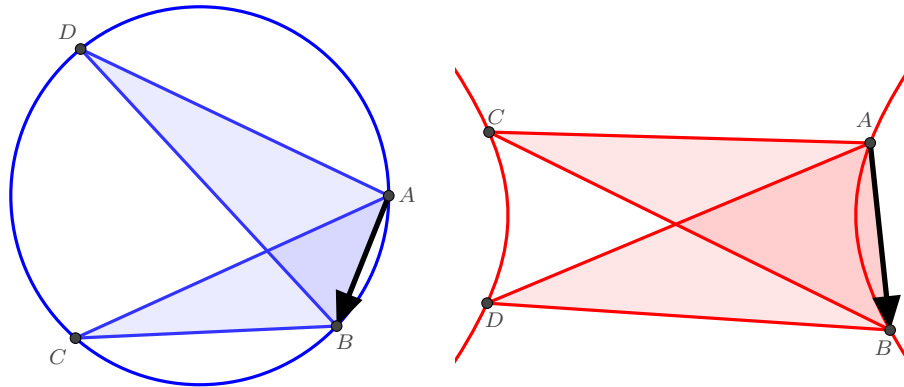
$$(C - A) \wedge (C - B) = (\alpha A + (\beta - 1)B) \wedge ((\alpha - 1)A + \beta B) = [(\alpha - 1)(\beta - 1) - \alpha\beta] A \wedge B$$

which simplifies to $(C - A) \wedge (C - B) = (1 - \alpha - \beta)A \wedge B$. Therefore we have

$$o(C - A, C - B) = \frac{(1 - \alpha - \beta)(A \cdot B + R^2)}{(1 - \alpha - \beta)A \wedge B} = \frac{A \cdot B + R^2}{A \wedge B}$$

which is independent of the position of C on our circle, as required.

By a change of coordinates, this statement applies to every triangle inscribed in a circle: the coturn at any vertex is completely determined by the circumquadrance and the opposite segment. Pictured below is the authors geometric visualization of this fact in the blue and red planes.



When dealing with angles, the equivalent fact is called the inscribed angle theorem, but it consists of two separate cases depending on if our point C is on the major arc or the minor arc. Technically the proof of the inscribed angle theorem requires three separate cases to

consider as well. With coturns, there is no casework, the formula valid for every part of a circle. With these two theorems in hand we can prove theorems that are out of reach for spreads and crosses of vertices.

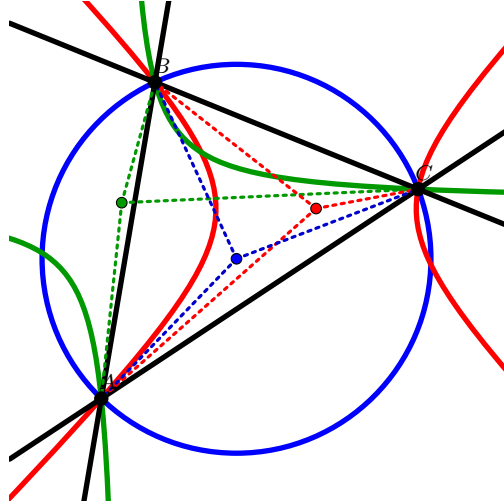
Exercise 2 So far we have shown that it is necessary for a cyclic quadrilateral to have pairs of opposite vertices with equal coturns. Show that it is also a sufficient condition to guarantee a quadrilateral is cyclic.

In exercise 5 of the essay Projection, Rejection, and Spread we showed that the circumcenter O and circumquadrance R^2 of triangle ABC with boundary $\mathbf{a} = C - B$, $\mathbf{b} = A - C$, $\mathbf{c} = B - A$ were

$$O = \frac{(\mathbf{b}^2 + \mathbf{c}^2 - \mathbf{a}^2)\mathbf{a}^2 A + (\mathbf{c}^2 + \mathbf{a}^2 - \mathbf{b}^2)\mathbf{b}^2 B + (\mathbf{a}^2 + \mathbf{b}^2 - \mathbf{c}^2)\mathbf{c}^2 C}{2(\mathbf{a}^2\mathbf{b}^2 + \mathbf{b}^2\mathbf{c}^2 + \mathbf{c}^2\mathbf{a}^2) - (\mathbf{a}^4 + \mathbf{b}^4 + \mathbf{c}^4)},$$

$$R^2 = \frac{\mathbf{a}^2\mathbf{b}^2\mathbf{c}^2}{2(\mathbf{a}^2\mathbf{b}^2 + \mathbf{b}^2\mathbf{c}^2 + \mathbf{c}^2\mathbf{a}^2) - (\mathbf{a}^4 + \mathbf{b}^4 + \mathbf{c}^4)}$$

so that the equation of the circumcircle (ABC) was $(X - O) = R^2$.



There is another way to find the equation of a circumcircle which can be more efficient to compute, when we have access to a basis.

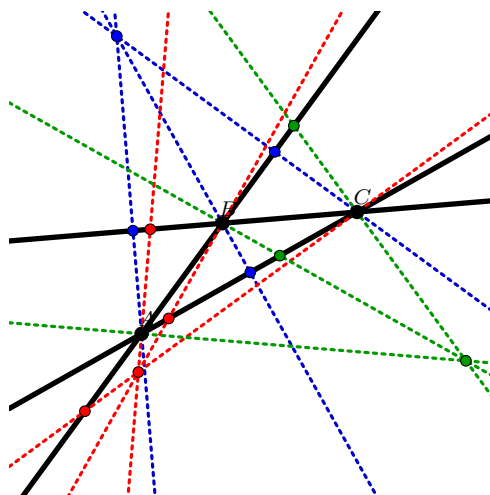
Lemma 1 Suppose that in a basis $\mathbf{e}_1, \mathbf{e}_2$ for the plane we have a vector $\mathbf{x} = x\mathbf{e}_1 + y\mathbf{e}_2$ where $\mathbf{x}^2 = ax^2 + 2bxy + cy^2$. Then the circumcircle of the points $P_i = (x_i, y_i)$ for $i = 1, 2, 3$ is given by the following determinant equation

$$\det \begin{bmatrix} ax^2 + 2bxy + cy^2 & x & y & 1 \\ ax_1^2 + 2bx_1y_1 + cy_1^2 & x_1 & y_1 & 1 \\ ax_2^2 + 2bx_2y_2 + cy_2^2 & x_2 & y_2 & 1 \\ ax_3^2 + 2bx_3y_3 + cy_3^2 & x_3 & y_3 & 1 \end{bmatrix} = 0.$$

Proof The Laplace expansion for determinants along the top row of this 4x4 matrix shows that this equation is quadratic in x, y and has the right quadratic form. Since this determinant has linearly dependent rows whenever we plug in any of the P_i 's, it must be the circle passing through those 3 points.

While the lemma above is not strictly necessary to do any of the following exercises, and in fact you should avoid manipulating any equation for a circumcircle in favor of using theorems 1 and 2, this lemma is helpful for graphically verifying the results of these exercises.

Recall from elementary geometry that an **Altitude** of triangle ABC from point A to side non-null BC is the unique line passing through A perpendicular to side BC . The **Foot** of an altitude is the point where it intersects the opposite side of the point it goes through. It can be proven that the three altitudes of a triangle always intersect at a single point called the **Orthocenter** H , but we will give a slick proof using Miquel's three circle theorem.



Pictured above is the blue, red and green altitudes and orthocenters of a triangle ABC . in 2008 Wildberger showed using direct computations that any colored orthocenter lies on the fourth intersection of the two circumcircles of the other colors, and that every colored circumcenter is the midpoint of the orthocenters of the other two colors. Thus the three colored orthocenters of a triangle ABC form what is known as their **Omega triangle**. The fact that the centroid of a triangle splits its the segment between its orthocenter and circumcenter in a 2:1 ratio means that ABC has the same centroid as its omega triangle. Additionally, he was able to show that given a point A and a side BC , any two colored altitudes are perpendicular from the perspective of the third color.

At this time it is not known if there is an elegant proof of these facts which explains the source of these magical coincidences rather than just verifies their correctness.

The next lemma is presented in the same spirit as the last, it is useful for graphically verifying the exercises, as well as for verifying wildberger's claims, but is too unwieldy to be satisfyingly deployed in proofs for what we have in mind when we have coturns at our disposal.

Lemma 2 Given a triangle ABC with boundary $\mathbf{a} = C - B$, $\mathbf{b} = A - C$, $\mathbf{c} = B - A$, let $h_A : X \cdot (B - C) = A \cdot (B - C)$, $h_B : X \cdot (A - B) = C \cdot (A - B)$, $h_C : X \cdot (C - A) = B \cdot (C - A)$ be the altitudes of the triangle. Then feet of the altitudes are given by

$$F_A = \frac{-(\mathbf{a} \cdot \mathbf{c})B - (\mathbf{a} \cdot \mathbf{b})C}{\mathbf{a}^2}, \quad F_B = \frac{-(\mathbf{b} \cdot \mathbf{c})A - (\mathbf{a} \cdot \mathbf{b})C}{\mathbf{b}^2}, \quad F_C = \frac{-(\mathbf{b} \cdot \mathbf{c})A - (\mathbf{a} \cdot \mathbf{c})B}{\mathbf{c}^2},$$

and the orthocenter is given by

$$H = \frac{(\mathbf{c}^2 + \mathbf{a}^2 - \mathbf{b}^2)(\mathbf{a}^2 + \mathbf{b}^2 - \mathbf{c}^2)A + (\mathbf{b}^2 + \mathbf{c}^2 - \mathbf{a}^2)(\mathbf{a}^2 + \mathbf{b}^2 - \mathbf{c}^2)B + (\mathbf{b}^2 + \mathbf{c}^2 - \mathbf{a}^2)(\mathbf{c}^2 + \mathbf{a}^2 - \mathbf{b}^2)C}{2(\mathbf{a}^2\mathbf{b}^2 + \mathbf{b}^2\mathbf{c}^2 + \mathbf{c}^2\mathbf{a}^2) - (\mathbf{a}^4 + \mathbf{b}^4 + \mathbf{c}^4)}.$$

Proof We will only derive the expressions for F_A and H , the logic is identical for F_B and F_C . The vector from A to a point on BC can be written in the form $\mathbf{c} + t\mathbf{a}$. In order for that vector to be on the altitude we need $\mathbf{a} \cdot (\mathbf{c} + t\mathbf{a}) = 0$ which implies $t = -\mathbf{a}^{-1} \cdot \mathbf{c}$. Then, using the fact that $\mathbf{a}^2 + \mathbf{a} \cdot \mathbf{c} = -\mathbf{a} \cdot \mathbf{b}$ we have that

$$A + \mathbf{c} - (\mathbf{a}^{-1} \cdot \mathbf{c})\mathbf{a} = B - (\mathbf{a}^{-1} \cdot \mathbf{c})\mathbf{a} = B - (\mathbf{a}^{-1} \cdot \mathbf{c})(C - B) = (1 + \mathbf{a}^{-1} \cdot \mathbf{c})B - (\mathbf{a}^{-1} \cdot \mathbf{c})C$$

and therefore $F_A = -(\mathbf{a}^{-1} \cdot \mathbf{b})B - (\mathbf{a}^{-1} \cdot \mathbf{c})C$ as required for the first part. Now we will assume familiarity with the affine theory of barycentric coordinates. If we have three points

$$X = \frac{bB + cC}{b + c} \quad Y = \frac{cC + aA}{c + a} \quad Z = \frac{aA + bB}{a + b}$$

then the lines AX , BY , CZ intersect at a point

$$W = \frac{aA + bB + cC}{a + b + c}.$$

Our goal will be to find a way of rewriting our previous formulas for the altitude in terms of three common parameters. All we need to do is multiply F_A by $(\mathbf{b} \cdot \mathbf{c})/(\mathbf{b} \cdot \mathbf{c})$, F_B by $(\mathbf{c} \cdot \mathbf{a})/(\mathbf{c} \cdot \mathbf{a})$, and F_C by $(\mathbf{a} \cdot \mathbf{b})/(\mathbf{a} \cdot \mathbf{b})$ to find the parameters we need, so that our three altitudes intersect at the point

$$H = \frac{(\mathbf{a} \cdot \mathbf{b})(\mathbf{a} \cdot \mathbf{c})A + (\mathbf{a} \cdot \mathbf{b})(\mathbf{b} \cdot \mathbf{c})B + (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{c})C}{-(\mathbf{b} \cdot \mathbf{c})\mathbf{a}^2 - (\mathbf{a} \cdot \mathbf{c})\mathbf{b}^2 - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}^2}$$

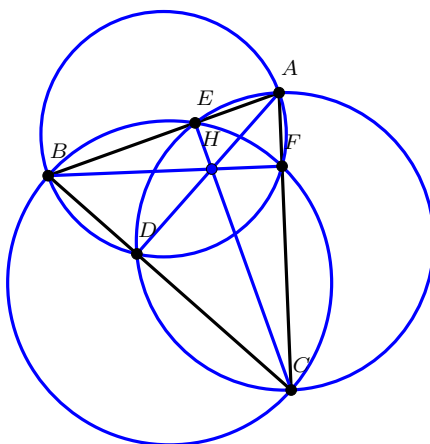
and the result follows from the facts that

$$\mathbf{a} \cdot \mathbf{b} = \frac{1}{2}(\mathbf{b}^2 - \mathbf{c}^2 - \mathbf{a}^2), \quad \mathbf{b} \cdot \mathbf{c} = \frac{1}{2}(\mathbf{a}^2 - \mathbf{b}^2 - \mathbf{c}^2), \quad \mathbf{c} \cdot \mathbf{a} = \frac{1}{2}(\mathbf{c}^2 - \mathbf{a}^2 - \mathbf{b}^2),$$

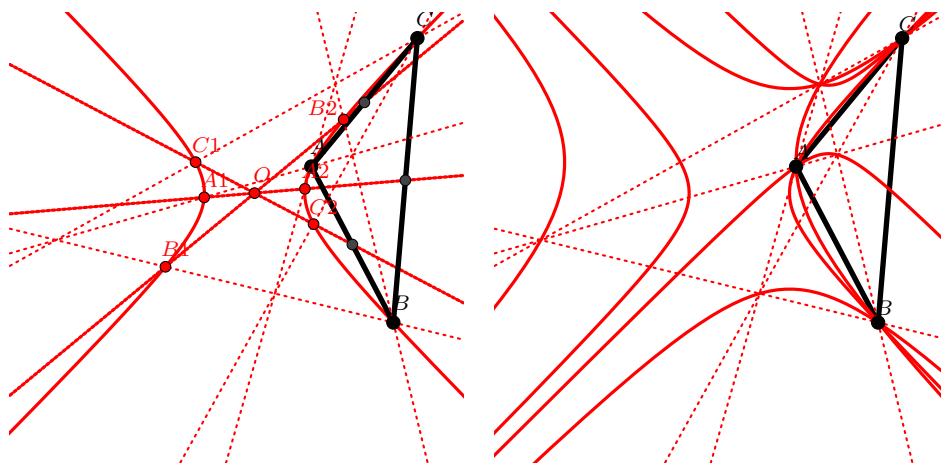
$$(\mathbf{b}^2 + \mathbf{c}^2 - \mathbf{a}^2)\mathbf{a}^2 + (\mathbf{c}^2 + \mathbf{a}^2 - \mathbf{b}^2)\mathbf{b}^2 + (\mathbf{a}^2 + \mathbf{b}^2 - \mathbf{c}^2)\mathbf{c}^2 = 2(\mathbf{a}^2\mathbf{b}^2 + \mathbf{b}^2\mathbf{c}^2 + \mathbf{c}^2\mathbf{a}^2) - (\mathbf{a}^4 + \mathbf{b}^4 + \mathbf{c}^4).$$

Now that we can graph everything to verify the claims that will be made, we can use coturns to demonstrate the existence of remarkable configurations of lines and circles.

Exercise 3 Given a triangle ABC with opposite feet of altitudes D, E, F respectively, and orthocenter H , show that $ADEB$, $BEFC$, and $CFDA$ are cyclic quadrilaterals.

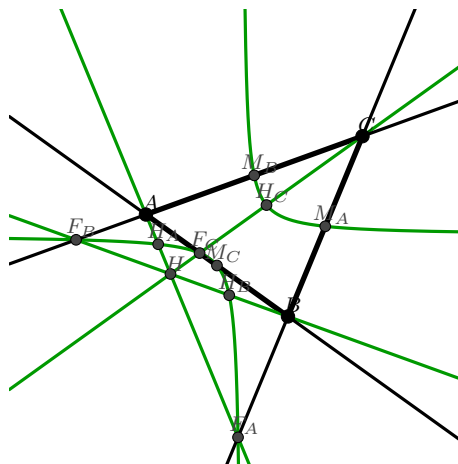


Exercise 4 Given a triangle ABC , let its perpendicular bisectors $p_{AB} : X \cdot (B - A) = (B^2 - A^2)/2$, $p_{BC} : X \cdot (C - B) = (C^2 - B^2)/2$, $p_{CA} : X \cdot (C - A) = X \cdot (C^2 - A^2)/2$ intersect the circumcircle (ABC) at points C_1 and C_2 , A_1 and A_2 , B_1 and B_2 respectively. Show that the six lines $AA_1, AA_2, BB_1, BB_2, CC_1, CC_2$ are vertex bisectors of the triangle ABC and form an orthic system of 4 points called consisting of opposite excenters E_A, E_B, E_C to ABC respectively, and incenter I . Conclude that $AIBE_C, BICE_A$, and $CIAE_B$ are all cyclic quadrilaterals.



The four points I, E_A, E_B, E_C are well known for being the centers of the four circles which are tangent to all three lines of the triangle ABC . In wildberger's book he gives formulas for the quadrances of these circles in terms of the spreads between the lines of the triangle and the six lines making up our vertex bisectors.

Exercise 5 Given triangle ABC with opposite feet of altitudes F_A, F_B, F_C respectively, circumcenter O , orthocenter H , midpoints of the perimeter $M_A = (B + C)/2$, $M_B = (C + A)/2$, $M_C = (A + B)/2$, and orthomidpoints $H_A = (H + A)/2$, $H_B = (H + B)/2$, $H_C = (H + C)/2$, show that the ninegon $DEFM_AM_BM_CH_AH_BH_C$ is cyclic. Call the circle that these points lie on the **ninepoint circle**, and show that the orthocenter is a center of dilation bringing the ninepoint circle onto the circumcircle. Conclude that the ninepoint center is $N = (H + O)/2$ the midpoint of the circumcenter and the orthocenter, and that the quadrance of the ninepoint circle is a quarter of the circumquadrance.



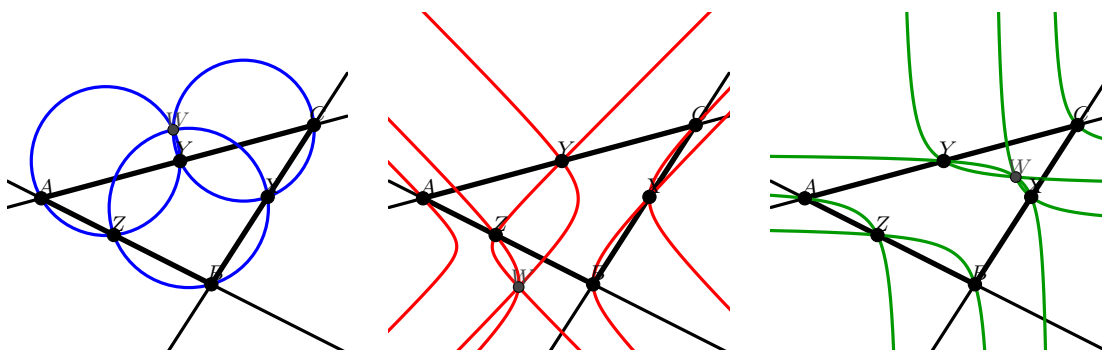
We will now prove two theorems of Miquel using the tools available.

Theorem 3 *Miquel's three circle theorem* Given a triangle ABC with points X, Y, Z on sides BC, CA, AB respectively, let W be the other intersection of circles (YAZ) and (ZBX) which is not Z . Then W is on circle (XCY) .

Proof Since $YAZW$ and $ZBXW$ are cyclic quadrilaterals then we have

$$o_A = o(A - Y, A - Z) = o(W - Y, W - Z), \quad o_B = o(B - Z, B - X) = o(W - Z, W - X).$$

Now we notice that $o_C = o(C - X, C - Y) = \frac{-(o_A + o_B)}{1 - o_A o_B}$, and that $o(W - X, W - Y) = \frac{-(o_A + o_B)}{1 - o_A o_B}$ meaning that $XCYW$ is a cyclic quadrilateral as required.



Exercise 6 Use Miquel's three circle theorem to deduce the existence of the orthocenter.

Exercise 7 Suppose that lines l_1, l_2, l_3, l_4 , intersect at points $P_{12} = l_1 \cap l_2, P_{13} = l_1 \cap l_3, P_{14} = l_1 \cap l_4, P_{23} = l_2 \cap l_3, P_{24} = l_2 \cap l_4, P_{34} = l_3 \cap l_4$. Show that the four circles $(P_{12}P_{13}P_{23}), (P_{12}P_{14}P_{24}), (P_{13}P_{14}P_{34}),$ and $(P_{23}P_{24}P_{34})$ all intersect a single point. Show that the centers of these four circles also form a cyclic quadrilateral.

